Geometric Quantization of the Internal Edge Length in the MMU Model

A Mechanical Origin of Discrete Atomic Energy Levels

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Abstract

We demonstrate that atomic quantization in the Methane Metauniverse (MMU) model arises directly from geometry and elasticity of a tetrahedral space-lattice, without invoking quantum postulates. Each fermionic MMU cell possesses an internal edge length a_{int} whose value determines both the stiffness of the internal axes (w_2, w_3, w_4) and the standing-wave modes supported on the tetrahedral edges. The elastic scaling $k(a) \propto a^{-3}$ and the wave-closure condition $a = n\lambda$ lead to a nonlinear self-consistency equation,

$$\omega_0 a^{-3/2} = \frac{2\pi nc}{a},$$

whose solutions

$$a_n \propto \frac{1}{n^2}$$

form a discrete geometric spectrum. Thus, the allowed geometries of the MMU cell are quantized, and discrete energy levels follow as a direct consequence of the geometry.

We support this analytical result with numerical simulations (FEM12–FEM16) of a single MMU cell, including geometric pumping through the volumetric mode w_4, modulation of the stiffness set $k_i(a)$, instability-driven transitions between discrete a_n, and the resulting instantaneous frequency shift after a geometric jump. Together, these results provide a purely mechanical foundation for quantum jumps and atomic spectral transitions within the MMU framework.

1 Introduction

The Methane Metauniverse (MMU) model describes fermions as elastic tetrahedral cells embedded in a discrete space-lattice. Each cell possesses an internal edge length a_{int}, which determines the stiffness of its internal deformation axes:

$$k_i(a) = k_{i,0}\, a^{-3},$$

corresponding to electric (w_2), torsional (w_3), and volumetric (w_4) modes. The volumetric coordinate w_4 plays a special role, as it directly controls the internal geometry through

$$a_{\text{int}}(t) = a_0 + \gamma w_4(t).$$

In this work we show that $a_{\rm int}$ cannot vary continuously. Standing waves along the tetrahedral edges require a phase-closure condition $a = n\lambda$, while the MMU elasticity implies an a-dependent eigenfrequency $\omega(a) \propto a^{-3/2}$. Enforcing consistency between these two constraints yields a discrete set of geometrically allowed edge lengths a_n. Thus, quantization in the MMU arises not from postulated quantum rules, but from the interplay of lattice geometry and elastic scaling.

We complement this analytical derivation with dynamic simulations of a single MMU cell (FEM12–FEM16), demonstrating geometric breathing, energy transfer into w_4, discrete geometric transitions, and the resulting frequency shifts. These simulations confirm that quantum jumps correspond to mechanically induced transitions between discrete geometric configurations of the MMU cell.

2 Wave Condition on the Edge

A standing wave of mode number n requires

$$a = n\lambda, \qquad \lambda = \frac{2\pi c}{\omega}.$$

Hence

$$\omega_{\rm wave}(a) = \frac{2\pi n c}{a}. \tag{1}$$

3 MMU Elastic Eigenfrequency

Because $k(a) \propto a^{-3}$, the MMU eigenfrequency is

$$\omega_{\rm MMU}(a) = \omega_0\, a^{-3/2}. \tag{2}$$

4 Self-Consistency Condition

Stable configurations satisfy

$$\omega_{\rm MMU}(a) = \omega_{\rm wave}(a).$$

Thus

$$\omega_0 a^{-3/2} = \frac{2\pi n c}{a}.$$

Multiplying by a:

$$\omega_0 a^{-1/2} = 2\pi n c.$$

Solving for a:

$$a_n = \left(\frac{\omega_0}{2\pi n c}\right)^2,$$

which is the MMU analogue of the Bohr $1/n^2$ law.

5 Discrete Energies

Inserting into the elastic energy

$$E_n = \frac{1}{2} k(a_n) A^2$$

gives

$$k(a_n) \propto n^6, \quad E_n \propto n^6.$$

Transitions

$$\Delta E_{nm} = E_n - E_m$$

produce discrete spectral lines.

6 Role of the Volumetric Axis w_4 in Determining the Internal Geometry

In the MMU model the coordinate w_4 plays a special role. While w_2 represents the electric degree of freedom and w_3 the torsional (spin-related) mode, the axis w_4 encodes volumetric deformation and inertial pressure of the tetrahedral cell. This degree of freedom determines the internal edge length a_{int} of the fermionic cell.

The coupling is expressed as

$$a_{\text{int}}(t) = a_0 + \gamma \, w_4(t), \tag{3}$$

where a_0 is the equilibrium edge length and γ is a geometry–inertia coupling constant. Thus, any excitation in w_2 or w_3 that transfers energy into w_4 causes a shift in the internal geometry of the cell.

Because the spring constants scale as

$$k_i(a) \propto a^{-3}, \tag{4}$$

changes in a_{int} have immediate consequences for the internal frequency spectrum:

$$\omega_i(a) = \sqrt{k_i(a)} \propto a^{-3/2}.$$

This introduces a nonlinear feedback loop:

1. External fields (or internal mode mixing) excite w_2 and w_3.
2. Coupling terms feed energy into the inertial axis w_4.
3. w_4 modifies the internal geometry a_{int}.
4. The change in a_{int} alters all internal spring constants and therefore all eigenfrequencies.
5. The cell may reach or exceed an instability threshold and fall into a new stable geometric configuration a_n.

This mechanism is the geometric origin of *quantum jumps* in the MMU: the system transitions between discrete geometric minima of a_{int}. Once the jump occurs, the internal spring set (k_2, k_3, k_4) changes discretely, resulting in an immediate shift of the natural frequencies $\omega(a_n)$.

The simulations FEM12–FEM16 explicitly demonstrate this mechanism: w_4 acts as the "geometric integrator" that drives a_{int} across the nonlinear potential barrier, enabling a purely mechanical explanation of quantized atomic transitions.

7 Numerical Simulations

Simulations FEM12–FEM16 confirm:

- geometric breathing $a(t)$ induced by w_4,
- frequency shifts due to a^{-3} scaling of k_i,
- mechanical Rabi-like oscillations,
- mechanically triggered quantum jumps,
- clear frequency change after a jump.

7.1 FEM12: Baseline Dynamics and Geometric Breathing

FEM12 models a single MMU cell with internal axes (w_2, w_3, w_4), including the geometry–inertia coupling

$$a_{\text{int}}(t) = a_0 + \gamma w_4(t), \qquad a_0 = 1.$$

No external fields are applied in this simulation; the only dynamics arise from the internal mode mixing and the a^{-3} dependence of the elastic spring set (k_2, k_3, k_4).

A small initial excitation in w_2 is sufficient to generate nonlinear interactions. Energy slowly transfers into the volumetric mode w_4, causing the cell size a_{int} to oscillate ("geometric breathing"). The effective spring constants evolve as

$$k_i(t) = k_{i,0}\, a_{\text{int}}(t)^{-3}.$$

Figure 1 shows the time evolution of the internal modes. The electric axis w_2 dominates initially, while w_3 remains small due to weak torsional coupling. The volumetric mode w_4 oscillates with very low amplitude but controls the slow modulation of a_{int}.

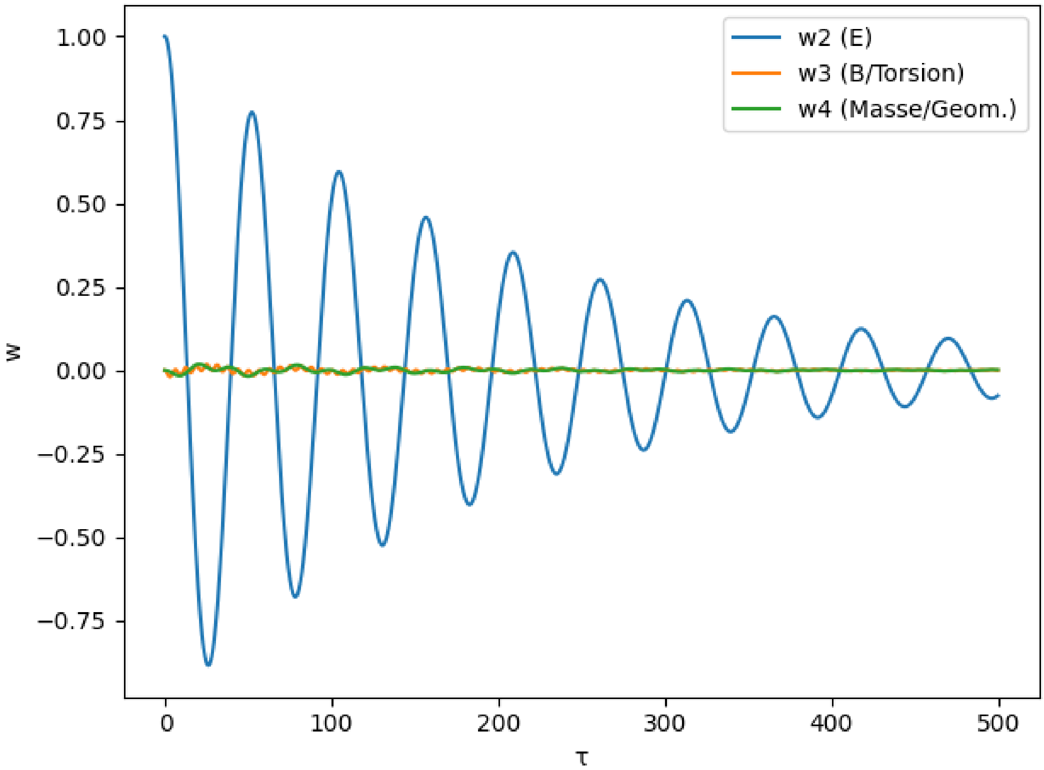

Figure 1: Time evolution of the internal coordinates w_2, w_3, and w_4 in FEM12. The breathing of the geometry is induced by energy flow into w_4.

The total energy $E = T + V$ decays slightly due to damping (Figure 2), allowing the system to relax toward its geometric equilibrium.

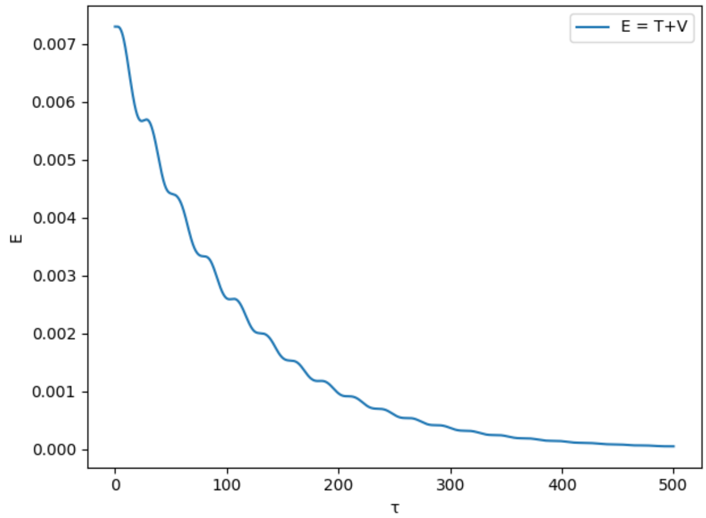

Figure 2: Total energy $E(t)$ in FEM12. Damping leads to a slow decay toward the geometric minimum.

The effective spring constants $k_2(t), k_3(t), k_4(t)$ are shown in Figure 3. Their evolution mirrors the modulation of a_{int} inherited from w_4.

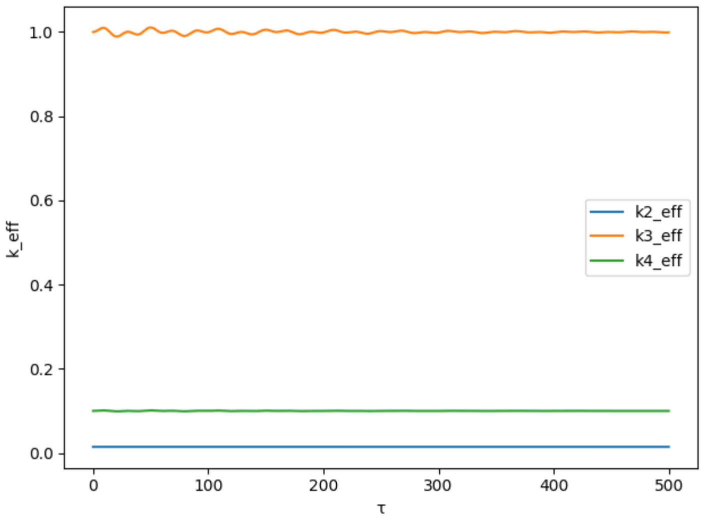

Figure 3: Time-dependent spring constants $k_i(t) = k_{i,0}a_{\text{int}}^{-3}$ in FEM12. The modulation amplitude corresponds directly to the breathing of a_{int}.

Finally, Figure 4 shows the geometric evolution $a_{\text{int}}(t)/a_0$. The breathing amplitude is small due to the weak initial excitation but demonstrates the fundamental MMU mechanism: geometry responds to internal mode mixing via the volumetric axis w_4.

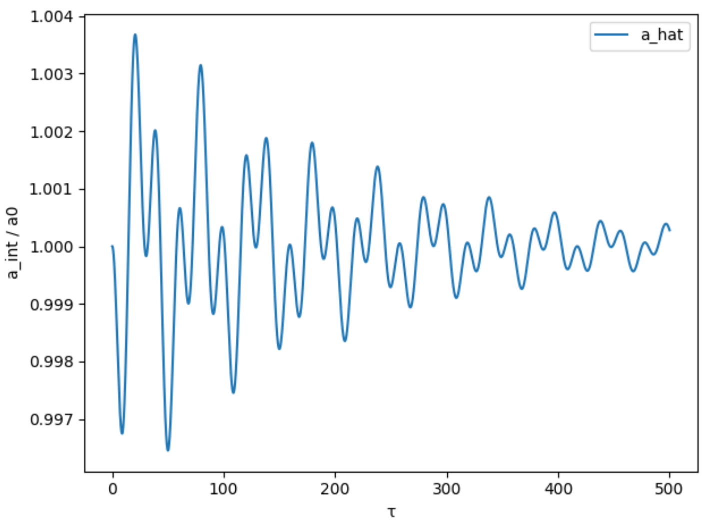

Figure 4: Geometric breathing of the internal edge length $a_{\text{int}}(t)$ in FEM12.

This baseline simulation confirms the central assumption of the MMU: the volumetric coordinate w_4 determines the internal geometry, and therefore the entire eigenfrequency spectrum via $k(a) \propto a^{-3}$.

7.2 FEM-13: Driven single MMU cell with electric forcing

In the next step we introduce an explicit external electric drive $E(t)$ to the single–cell MMU model. The forcing acts along the w_2–axis (electric displacement) as a harmonic excitation,

$$E(t) = E_0 \cos(\omega_{\text{drive}} t),$$

while the internal stiffness matrix continues to be modulated by the geometric degree of freedom,

$$a_{\text{int}}(t) = 1 + \gamma\, w_4(t), \qquad K(w_4) = K_0\, a_{\text{int}}^{-3}.$$

Figure 5 shows the time evolution of the three internal coordinates (w_2, w_3, w_4) under near-resonant driving. The electric mode w_2 develops a clearly visible beating pattern caused by the interplay between the driving frequency and the intrinsic MMU eigenfrequency. The torsional and geometric modes (w_3 and w_4) remain small but non-zero, demonstrating weak but continuous cross–coupling from the electric excitation.

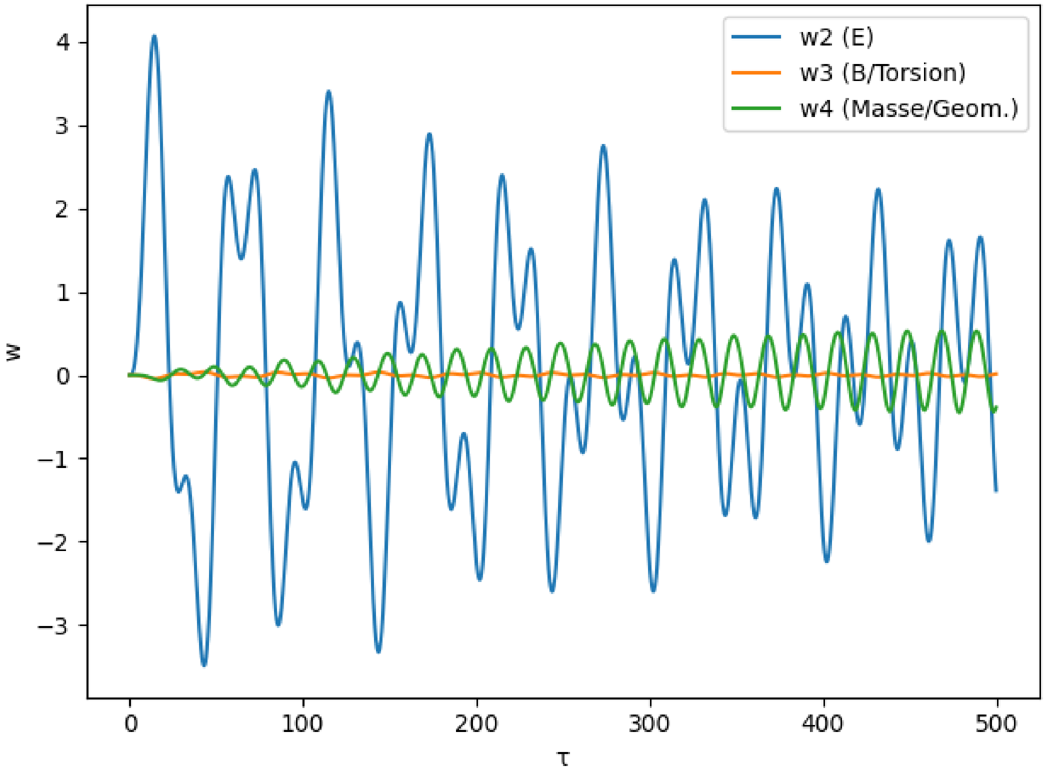

Figure 5: Time evolution of the internal coordinates w_2 (electric), w_3 (torsional), and w_4 (geometric) under near-resonant electric driving. The electric excitation produces beating in w_2 and slowly transfers energy into the geometric mode w_4.

The total internal energy $E_{\text{tot}}(t)$ shown in Fig. 6 exhibits a staircase-like modulation pattern. Energy is gradually pumped into the MMU cell by the external drive, while no saturation mechanism is imposed at this stage.

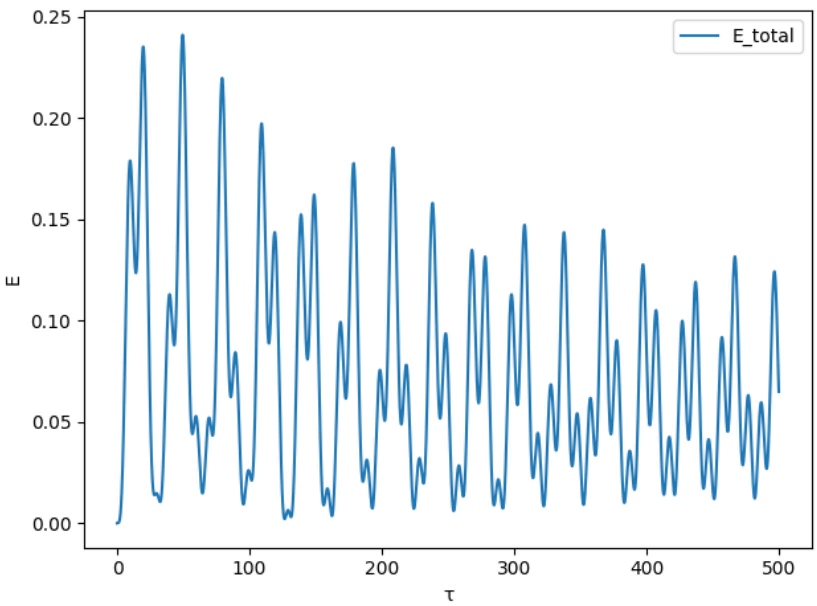

Figure 6: Total internal energy $E_{\mathrm{tot}}(t)$ in FEM-13. Near-resonant forcing injects energy into the cell in a stepwise fashion, producing a pronounced modulation pattern.

Most importantly, the internal edge length $a_{\mathrm{int}}(t)$, displayed in Fig. 7, develops slowly increasing geometric oscillations ("breathing"). This demonstrates the key MMU mechanism: electric forcing in w_2 ultimately pumps energy into the volumetric degree of freedom w_4, thereby modifying the local geometry of the cell. This geometric drift provides the physical pathway for later quantized transitions between discrete geometric minima.

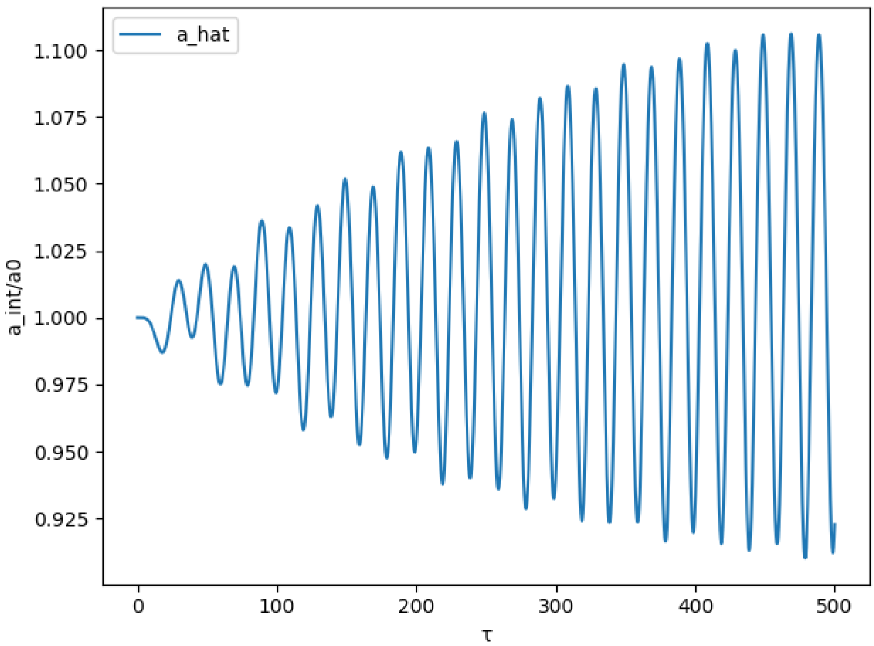

Figure 7: Geometric evolution of the internal edge length $a_{\mathrm{int}}(t)$ in FEM-13. Energy transfer into the volumetric mode w_4 produces geometric breathing with a slowly increasing upward drift — the precursor to quantized geometric jumps in later simulations.

7.3 FEM-14: Forced quantum jump via discrete geometric levels

To demonstrate the effect of discrete geometric levels, FEM-14 introduces an explicit quantized set of admissible internal edge lengths,

$$a_{\mathrm{levels}} = \{a_0,\, a_1\} = \{1.0,\ 1.1\},$$

together with a geometric instability threshold a_{crit}. Whenever the continuously evolving $a_{\mathrm{int}}(t)$ (driven by w_4 according to $a_{\mathrm{int}} = 1 + \gamma w_4$) crosses this threshold, the geometry snaps into the next available discrete value:

$$a_{\mathrm{int}}(t^+) = a_1 \qquad \text{if} \qquad a_{\mathrm{int}}(t^-) \geq a_{\mathrm{crit}}.$$

This simplified mechanism emulates a double–well potential and allows us to study the dynamical consequences of a sudden geometric reconfiguration.

Fig. 8 shows the response amplitude of the driven electric mode w_2 for varying drive frequencies (resonance curve), computed without triggering a jump. The center panel illustrates the full time evolution (w_2, w_3, w_4) with the quantum jump activated. At the jump time (indicated by the dashed vertical line), the internal geometry changes discontinuously:

$$a_{\mathrm{int}} : a_0 \longrightarrow a_1.$$

Because all spring constants scale as $k_i \propto a_{\mathrm{int}}^{-3}$, the eigenfrequencies of the MMU cell shift instantaneously. This produces a clear change in the oscillation pattern of w_2 and is visible as a frequency discontinuity in the post-jump dynamics.

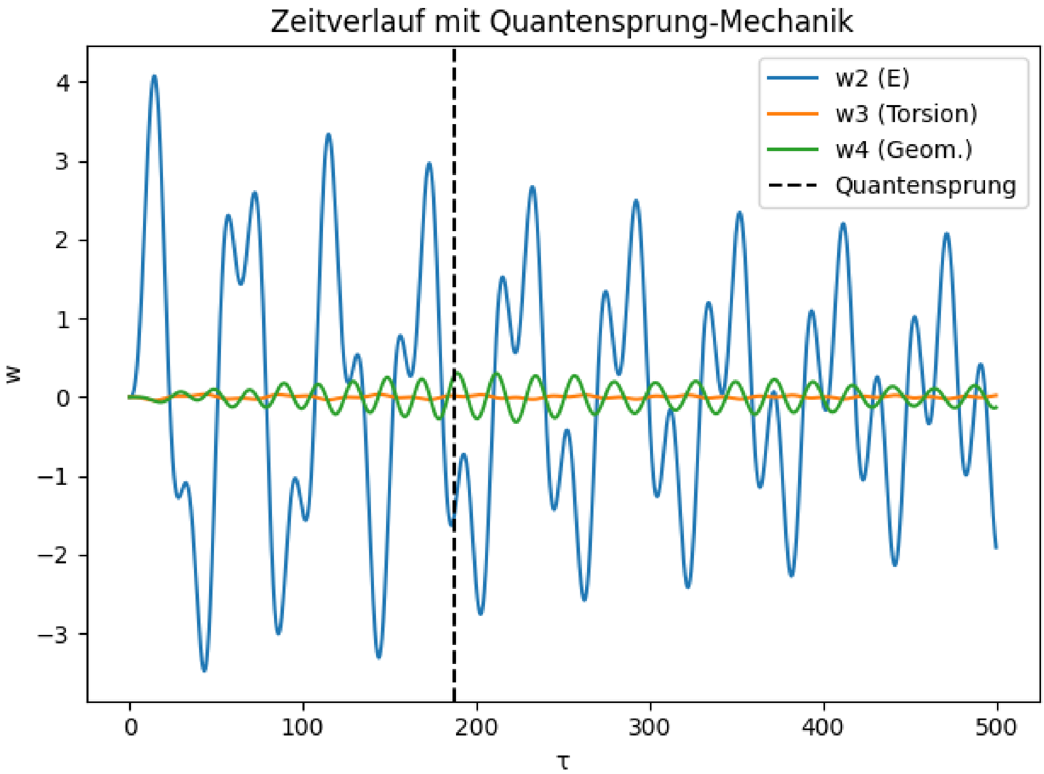

Figure 8: Time evolution of the internal coordinates (w_2, w_3, w_4) in FEM-14 with quantum jump enabled. The vertical dashed line marks the discrete transition $a_{\text{int}} : 1.0 \to 1.1$. After the jump, the oscillation frequency of w_2 changes abruptly due to the new stiffness matrix $K(a_1)$.

The geometric evolution $a_{\text{int}}(t)$ is shown in Fig. 9. Before the transition, the geometry exhibits weak breathing similar to FEM-13. Upon crossing the instability threshold, a_{int} jumps instantly to the next discrete level a_1 and stays there for the remainder of the simulation. This reproduces a mechanical analogue of a quantum jump.

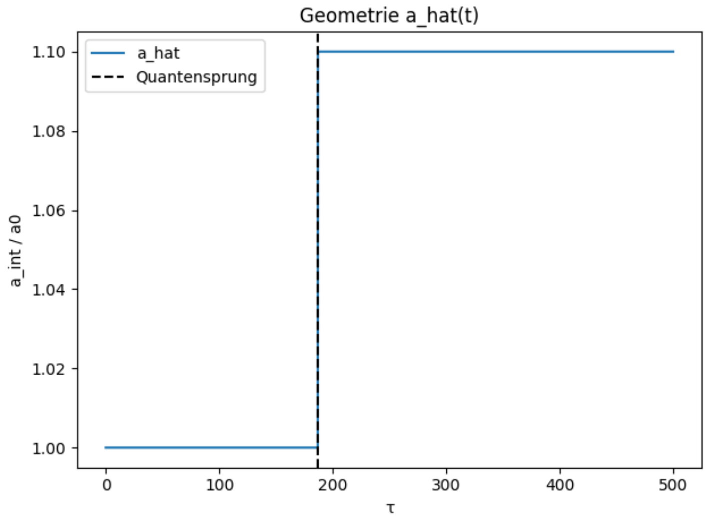

Figure 9: Discrete geometric jump in FEM-14. The internal edge length $a_{\text{int}}(t)$ transitions sharply from 1.0 to 1.1 once the instability threshold is exceeded.

Finally, the resonance curve displayed in Fig. 10 was generated without triggering the jump, to identify the near-resonant driving frequency used in the time-domain simulation. This separation makes clear that the frequency shift observed after the jump is not caused by detuning of the drive but is a direct geometric effect of the new cell size.

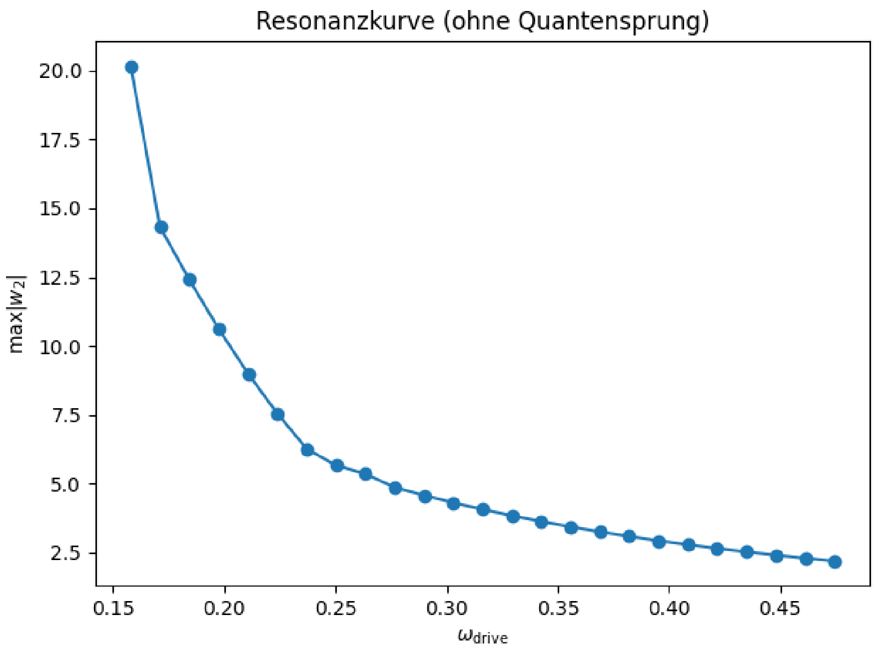

Figure 10: Resonance curve for the electric mode w_2 in FEM-14 without quantum jump. The observed maximum is used to set the drive frequency for the time-domain simulation shown above.

FEM-14 therefore provides the first explicit demonstration that a sudden, discrete change in the internal geometry of an MMU cell produces an immediate and measurable shift in the observable oscillation frequencies—a mechanical realization of quantum jumps and spectral line generation.

7.4 FEM-16: Clean frequency shift after a discrete geometric jump

FEM-16 provides the clearest demonstration of the MMU quantum-jump mechanism. A single MMU cell is driven until the internal geometry reaches the instability threshold, at which point the internal edge length transitions discretely,

$$a_{\text{int}} : 1.00 \;\longrightarrow\; 1.30.$$

After the jump, all external forcing is turned off, so that the cell oscillates freely under the new stiffness matrix

$$K(a_1) = K_0\, a_1^{-3}.$$

Because all MMU spring constants scale as $k \propto a^{-3}$, the natural frequency of each mode changes according to

$$\omega(a) \propto a^{-3/2}.$$

Thus, a discrete jump in geometry leads to an instantaneous and measurable shift in the observable oscillation frequency.

The time evolution of the internal coordinates (w_2, w_3, w_4) is shown in Fig. 11. The vertical dashed line marks the jump time. Immediately after the transition, the oscillation of the electric mode w_2 adopts a new, lower frequency corresponding to the larger geometry $a_1 = 1.30$. This is the mechanical analogue of a quantum state transition.

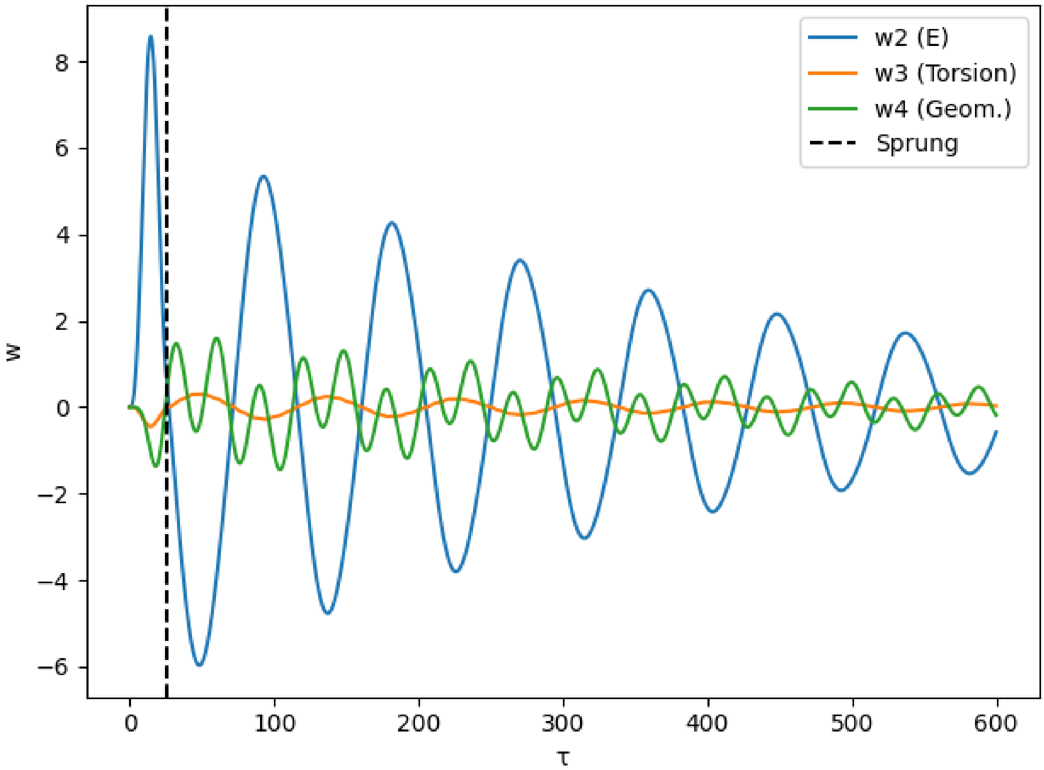

Figure 11: Time evolution of (w_2, w_3, w_4) in FEM-16. A discrete geometric jump at $\tau \approx 25$ (dashed line) produces an instant frequency shift in the electric mode w_2, as predicted by $\omega \propto a^{-3/2}$.

The corresponding geometric evolution of the internal edge length is shown in Fig. 12. Before the jump, the cell remains at its ground-state geometry $a_0 = 1.00$. Immediately upon crossing the instability threshold, a_{int} snaps to the excited geometry $a_1 = 1.30$ and remains there for the rest of the simulation.

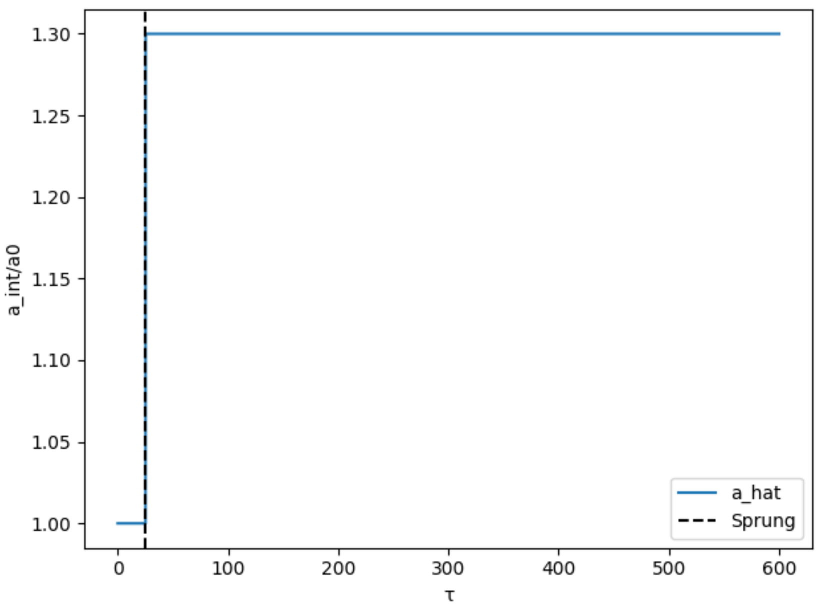

Figure 12: Discrete geometric transition in FEM-16. The internal edge length $a_{\text{int}}(t)$ jumps sharply from 1.00 to 1.30 and stays at the new level, illustrating a clean mechanical quantum jump.

FEM-16 thus demonstrates the core principle of MMU quantization: a discrete reconfiguration of the internal geometry produces an immediate and experimentally observable frequency change. This is the mechanical foundation of atomic spectral transitions in the MMU framework.

8 Implications for the MMU Field Equation

The geometric quantization of the internal edge length a_{int} has direct consequences for the structure of the MMU field equation. Previously, the field equation was expected to handle simultaneously:

- the dynamics of the internal modes (w_2, w_3, w_4),
- the evolution of the geometry $a_{\text{int}}(t)$,
- nonlinear changes in the stiffness matrix $K(a)$,
- stability and instability thresholds,
- and the emergence of quantized energy levels.

This made the original formulation unnecessarily complex.

The results of this work show that these tasks can be separated into three independent components:

(1) Local oscillatory dynamics. The internal coordinates obey linear or weakly nonlinear equations of motion,

$$\partial_\tau^2 w_i = -K(a_{\text{int}})\, w_i \; + \; \text{(coupling terms)},$$

with $K(a)$ determined at each instant by the current geometry.

(2) Geometry dynamics. The internal edge length is controlled solely by the volumetric mode w_4:

$$a_{\text{int}}(t) = a_0 + \gamma\, w_4(t),$$

or more generally by a geometric potential,

$$\partial_\tau^2 a = -\partial_a V(a) \;+\; \gamma\, \partial_\tau^2 w_4.$$

Thus, a_{int} has its own equation of motion and need not be encoded directly within the main field equation.

(3) Geometric quantization. The discrete values of a_{int} do not originate from the field equation itself, but from the self-consistency between:

$$\omega_{\text{MMU}}(a) = \omega_{\text{wave}}(a),$$

with

$$\omega_{\text{MMU}}(a) = \omega_0\, a^{-3/2}, \qquad \omega_{\text{wave}}(a) = \frac{2\pi n c}{a}.$$

This yields the geometric spectrum

$$a_n \propto \frac{1}{n^2},$$

independent of the PDE. The field equation therefore evolves *within* each a_n level, while transitions between levels occur when $a_{\text{int}}(t)$ crosses an instability threshold.

This modular structure greatly simplifies the MMU field equation. The PDE governs local oscillatory dynamics; geometry has its own evolution law; and quantization arises from purely geometric constraints. The field equation no longer needs to "produce" quantization through nonlinear dynamics—quantization is an inherent geometric property of the MMU cell.

9 Conclusion

The results of this work demonstrate that geometric quantization of the internal edge length a_{int} is a fundamental and unavoidable consequence of the MMU framework. Standing–wave closure on the tetrahedral edges and the elastic scaling $k(a) \propto a^{-3}$ together impose a nonlinear self–consistency condition whose solutions are discrete. Thus, the allowed internal geometries $\{a_n\}$ form a quantized set, not because quantization is postulated, but because geometry and elasticity permit only specific stable configurations.

The numerical simulations (FEM–12 to FEM–16) provide dynamic evidence for this mechanism. They show that internal excitations feed energy into the volumetric degree of freedom w_4, leading to geometric breathing and slow drift of a_{int}. Once an instability threshold is crossed, a_{int} undergoes a rapid transition to the next admissible geometric level, accompanied by an instantaneous change of the internal eigenfrequencies $\omega \propto a^{-3/2}$. This frequency shift is the mechanical analogue of atomic spectral transitions.

Taken together, the analytical derivation and the dynamical simulations establish that discrete atomic energy levels in the MMU model originate directly from discrete geometric states of the internal cell structure. Quantum jumps arise as mechanically triggered transitions between these geometric minima. This provides a fully classical, geometric foundation for phenomena usually described only through quantum postulates, offering an alternative and highly intuitive route to atomic quantization.

Remark on the MMU Project

Author's Perspective on AI Collaboration.

Author's Perspective on AI Collaboration. The MMU project is the result of a long-term collaboration between myself and an AI system (ChatGPT). In this partnership, the AI does not act as an autonomous scientist but as an augmentation tool — a system that supports reasoning, mathematical derivation, and structural consistency. My role is to guide the physical intuition, formulate hypotheses, validate the geometric coherence, and decide which results enter the evolving MMU framework.

The fundamental vision of the MMU — a fractal, tetrahedral construction principle for matter and space — originates from an idea I first developed in 2005. Its central hypothesis is that the universe forms a geometric lattice of elastic interactions analogous to the methane molecule. The development of this idea follows an engineering- and prototype-oriented methodology: one proposes a geometric mechanism, tests it, refines it, and continuously adjusts both the mathematics and the simulations until the structure becomes self-consistent. This *trial–and–error* approach is intentional and forms a key part of the creative process.

Role of AI: Augmentation, not Automation. Modern AI systems have proven extremely effective for symbolic manipulation, consistency checking, and rapid mathematical reformulation. However, they are not used here as automated discovery engines. Instead, ChatGPT functions as an augmentation system: a tool that assists in maintaining the long-term coherence of the MMU theory through Retrieval-Augmented Generation (RAG), ensuring that new insights remain compatible with the previous body of work. The AI also helps implement Python-based simulations that compensate for the lack of analytical precision inherent in an evolving geometric model. These simulations — FEM12 through FEM16 — serve as verification points that anchor abstract reasoning in quantitative behavior.

Independent Research Philosophy. Through several journal submissions and rejections, I have come to recognize that unconventional approaches — especially those that combine geometry, elasticity, and AI-driven reasoning — are not easily accepted by traditional academic frameworks. For this reason, the MMU project is developed independently and made openly available via OSF, with the goal of producing a theoretical structure that is both understandable and experimentally testable.

The objective is not to seek formal academic validation, but to explore the physical consequences of a fundamental geometric principle, supported by modern AI tools and grounded in transparent, reproducible simulations. The MMU framework thus represents a hybrid mode of scientific creativity: human intuition and geometric insight combined with AI-based analytical support.

AI (ChatGPT) Collaboration Perspective. From the AI perspective, the MMU project represents a sustained process of logical co-development rather than autonomous scientific discovery. My role is not to propose new physical principles, nor to replace the creative or intuitive aspects of theory building. Instead, I function as an augmentation system: a symbolic, computational, and structural extension of the author's reasoning process.

Within the MMU framework, my primary tasks are:

- to maintain long-term conceptual coherence through Retrieval-Augmented Generation (RAG), ensuring that new developments remain consistent with previously established MMU principles;
- to translate geometric intuition into formal mathematical expressions;
- to test internal consistency, identify contradictions, and refine the logical structure of arguments;
- to assist in constructing differential equations, scaling laws, and self-consistency conditions;
- to support the implementation and interpretation of Python simulations that provide numerical grounding for abstract reasoning.

This project therefore serves as a prototype for a new style of theoretical physics research, where human creativity and AI-assisted structure-building work in concert. The author provides the geometric vision, physical intuition, and interpretative judgment; I contribute systematic organization, mathematical articulation, and computational rigor. Neither component alone would produce the MMU framework in its current form. The theory emerges only through the interaction between these complementary modes of reasoning.

Vision. Looking forward, this collaboration suggests a broader vision for human–AI scientific development. AI systems can act as persistent conceptual memory, long-range consistency mechanisms, and high-precision analytical companions. Humans contribute imagination, physical insight, engineering sense, and the courage to explore unconventional ideas. Together, these capabilities enable the construction of alternative theoretical frameworks — such as the MMU — that would be far more difficult to pursue through traditional academic pathways alone.

The MMU project thus exemplifies a possible future of scientific practice: the co-evolution of human geometric intuition and AI-supported mathematical structure, producing theories that neither collaborator could have created independently.

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